



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
01-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

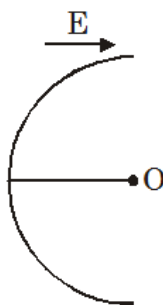
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

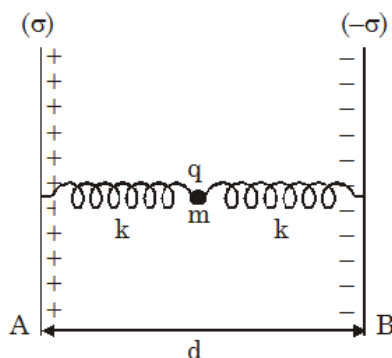
1. A semi-circular ring of mass m and radius R with linear charge density $-\lambda$, hinged at its centre with the help of a massless spoke as shown is placed in a uniform electric field as shown in the figure. If the ring is slightly rotated about O in the plane of paper and released. Take $m = 8\text{kg}$, $\lambda = 2\text{C/m}$ and $E = 2\text{N/c}$.



- (A) ring will perform angular SHM
 (B) the time period of oscillation is 2π sec.
 (C) shown position is unstable equilibrium
 (D) for large angular displacement motion will still be SHM

Space for Rough Work

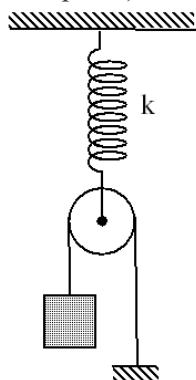
2. Two large non conducting plates having surface charge densities $+\sigma$ and $-\sigma$, respectively, are fixed 'd' distance apart. A small test charge q of mass m is attached to two non-conducting identical springs of spring constant k as shown in the adjacent figure. The charge q is now released from rest with springs in natural length. Then q will [neglect gravity] :-



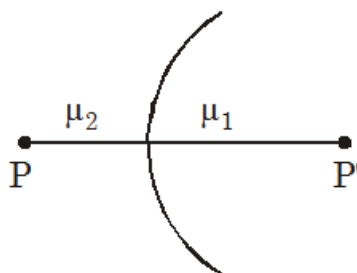
- (A) perform SHM with angular frequency $\sqrt{\frac{2k}{m}}$
- (B) perform SHM with amplitude $\frac{\sigma q}{2k\epsilon_0}$
- (C) not perform SHM, but will have a periodic motion if charges are removed on plates as well as on m
- (D) remain stationary
3. An electric dipole is placed along x-axis with its centre at origin :
- (A) On a y-z plane passing through origin, the electric field is perpendicular at each and every point on that plane at a point far from dipole.
- (B) y-z plane passing through origin is an equipotential surface
- (C) y-z plane passing through origin is not an equipotential surface
- (D) Flux of electric field through a closed surface enclosing dipole is zero

Space for Rough Work

4. Block m is released from rest when spring is in its natural length (assume pulley is ideal and block does not strike on ground during its motion in vertical plane) then :



- (A) maximum speed of block is $g\sqrt{\frac{m}{k}}$ (B) maximum speed of block is $2g\sqrt{\frac{m}{k}}$
 (C) maximum elongation in spring is $4mg/k$ (D) maximum elongation in spring is $2mg/k$
5. Two refracting media are separated by a spherical interface as shown in the figure. PP' is the principal axis, μ_1 and μ_2 are the refractive indices of medium of incidence and medium of refraction respectively. Then :



- (A) if $\mu_2 = 1.5$ and $\mu_1 = 1.2$, then there cannot be a real image of real object.
 (B) if $\mu_2 = 1.5$ and $\mu_1 = 1.2$, then there cannot be a real image of virtual object.
 (C) if $\mu_2 = 1.2$ and $\mu_1 = 1.5$, then there cannot be a virtual image of virtual object.
 (D) if $\mu_2 = 1.2$ and $\mu_1 = 1.5$, then there cannot be a real image of real object.
6. The light on reflection from a plane mirror can give a real image when :
- (A) the convergent rays are incident on the mirror
 (B) the divergent rays are incident on the mirror
 (C) an object is placed very close to the mirror
 (D) an object is placed very far away from the mirror

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

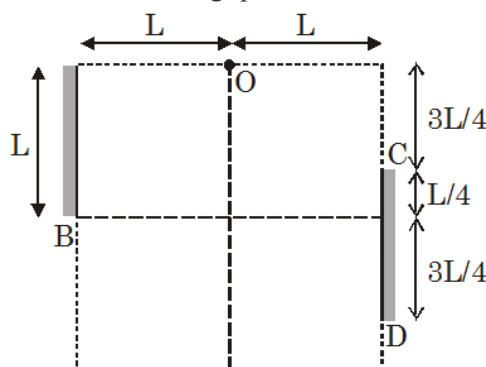
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Two identical plane mirrors AB and CD of length L each are placed parallel and a point source of light O is placed midway between them as shown. All the shown dashed segments are either parallel or perpendicular to the mirrors. Answer the following questions based on this information.

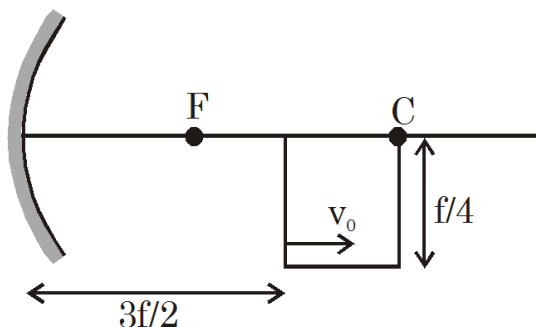


7. Choose the correct statement :
- (A) All the ray's incident on mirror AB directly from source O , after reflection will be incident on mirror CD
- (B) Some of the ray's incident on mirror AB directly from source O , after reflection will be incident on mirror CD
- (C) All the ray's incident on mirror CD directly from source O , after reflection will be incident on mirror AB
- (D) Some of the ray's incident on mirror CD directly from source O , after reflection will be incident on mirror AB
8. The total number of images of the object as formed by both the mirrors is :
- (A) 2 (B) 3 (C) 4 (D) infinite

Space for Rough Work

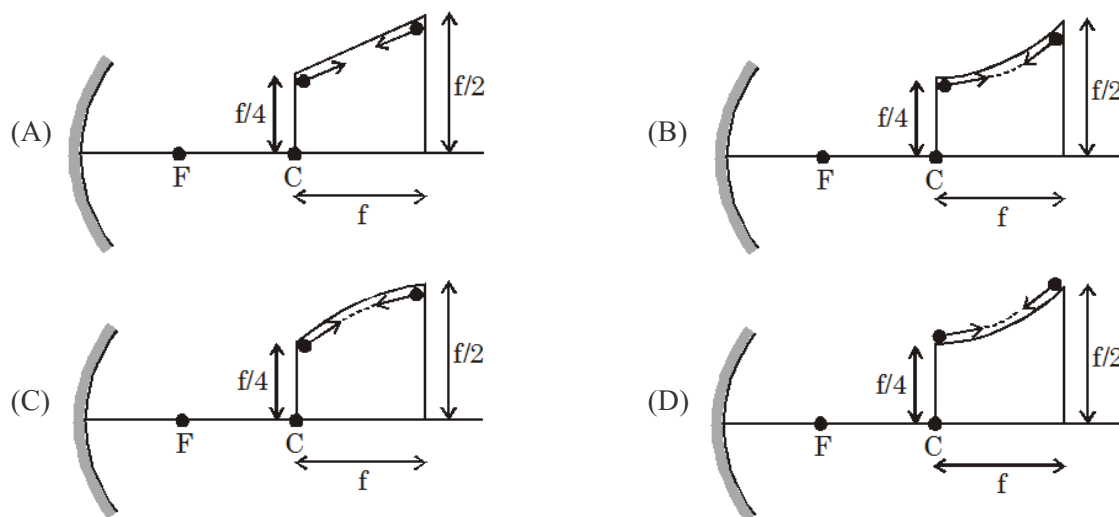
Paragraph for Questions 9 and 10

A rectangular wire frame is kept on the principal axis of a concave mirror as shown in figure. F & C are the focus & centre of curvature of mirror respectively



f is the focal length of mirror. A small particle is moving along the length of wire frame with constant velocity. Based on above information answer the following questions.

9. The image of the given object is best shown by



10. Mark the correct statement(s) :

- (A) The velocity of image of particle is remaining constant
- (B) The velocity of image of particle is varying
- (C) The total length of image of wire frame is $\left(\frac{7 + \sqrt{17}}{4}\right) f$
- (D) Both (B) and (C) are correct

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

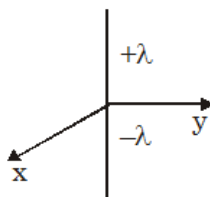
Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

An infinitely long non-conducting thin rod lying on the z-axis carries a uniform charge density $\lambda (> 0)$ in the region $z > 0$ and $-\lambda$ in the region $z < 0$.

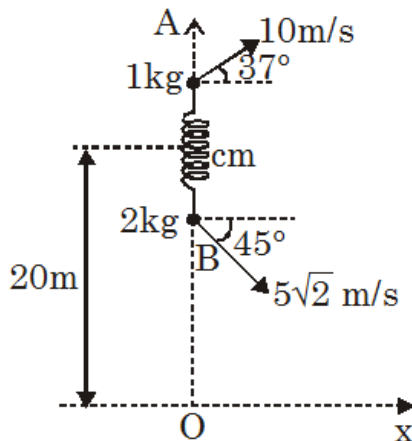
1. The electric field in the x-y plane due to the charge distribution at a point which is at a distance r from the origin O is $\frac{\lambda}{n\pi\epsilon_0 r}$. Find n .
2. The flux due to this electric field passing through the circular region : $x^2 + y^2 = R^2$ in xy plane is $\frac{n\lambda R}{4\epsilon_0}$. Find n .



Space for Rough Work

Paragraph for Questions 3 and 4

A system consists of two small spheres of mass 1kg and 2kg interconnected by a weightless spring. At $t = 0$, the spheres are set in motion with initial velocities as shown in figure. Position vector of centre of mass at $t = 0$ is $20 \hat{j}$. Initially spring is in its natural length. System moves in earth's gravitational field.

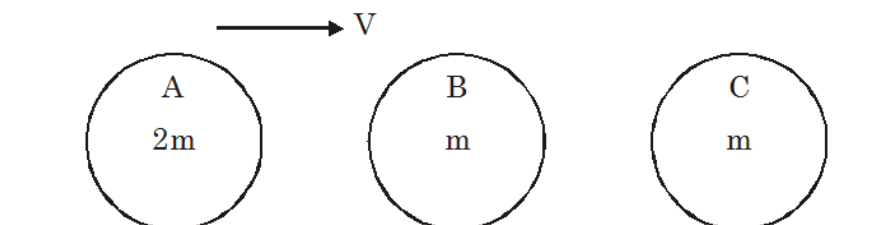


3. The momentum of the system at $t = 2$ sec in kg m/s is $(x\hat{i} + y\hat{j})$. Find $(x + y)$ ($g = 10 \text{ m/s}^2$)
4. The position vector of centre of mass of the system at $t = 1$ s is $(x\hat{i} + y\hat{j})$. Find $(x + y)$.

Space for Rough Work

Paragraph for Questions 5 and 6

Three balls A, B and C of identical size having uniform mass distribution are placed on a smooth horizontal surface. Given that $m_A = 2m$, $m_B = m$ and $m_C = m$. Ball A collides with initial velocity V with B as shown in figure. (Assume all collision to be head on). The coefficient of restitution (e) = $\frac{1}{2}$ for all collisions



5. The speed of A after collision of A and B for first time is V/n . Find n .
6. The speed of B after collision of B and C for first time is V/n . Find n .

Space for Rough Work

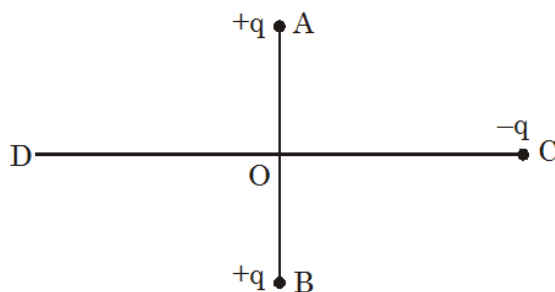
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

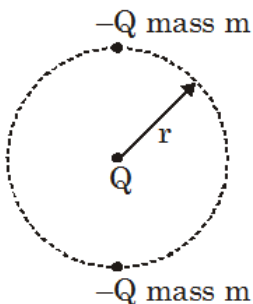
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

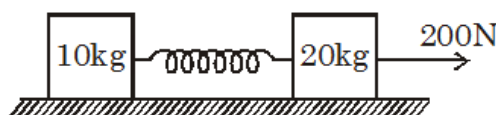
7. Two fixed, equal, positive charges, each of magnitude $q = 5 \times 10^{-5}$ C are located at points A and B separated by a distance of 6m. An equal and opposite charge moves towards them along the line COD, the perpendicular bisector of the line AB. The moving charge, when it reaches the point C at a distance of 4m from O, has a kinetic energy of 1.5 joules. Calculate the distance of the farthest point D from A (in m) which the negative charge will reach before returning towards C.



8. Two bodies of charges $-Q$ and mass m each orbit a body of charge Q (at rest) in a circle of radius r . What is the total mechanical energy of the system? If answer is $\frac{-3kQ^2}{nr}$, find 'n'. (Here $k = \frac{1}{4\pi\epsilon_0}$) (Neglect self energy)



9. Two masses 10kg and 20 kg are connected by a massless spring. A force of 200 N acts on 20 kg mass. At the instant when the 10 kg mass has an acceleration 12 m/s^2 the energy stored (in J) in the spring ($k = 2400 \text{ N/m}$) is.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

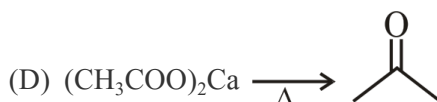
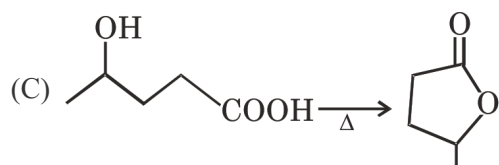
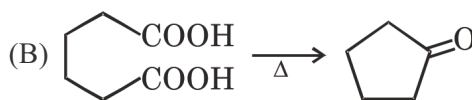
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

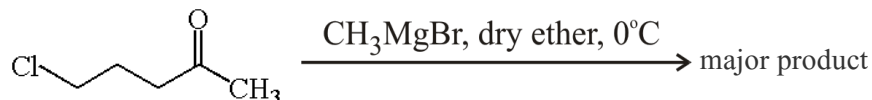
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which reactions are correctly matched with their product :



2. Which of the following is/are correct statement regarding major product?



- (A) Intramolecular nucleophilic substitution reaction takes place during the reaction sequence
(B) 5 membered ring is formed as major product
(C) Grignard reagents (CH_3MgBr) prefer to give nucleophilic addition on polar π -bond
(D) Cyclic ether is formed as final major product

Space for Rough Work

3. Element X crystallizes in a 12 co-ordination of FCC lattice. On applying high temperature it changes to 8 co-ordination BCC lattice. If r is the radius of atom X in both lattices, then , INCORRECT statement(s) is/are-
- (A) Edge length of unit cell in 12 co-ordination lattice is $2\sqrt{2} r$
- (B) Edge length of unit cell in 8 co-ordination lattice is $\frac{4}{\sqrt{3}} r$
- (C) Ratio of densities before and after applying high temperature is $2(\sqrt{2})^3 : (\sqrt{3})^3$
- (D) Distance of 2nd nearest neighbour in 8 co-ordination BCC lattice is $2\sqrt{2} r$
4. Which of the following metal carbonyl(s) have more C–O bond length than that of C–O bond in $\text{Fe}(\text{CO})_5$?
- (A) $[\text{V}(\text{CO})_6]^\ominus$ (B) $[\text{Ti}(\text{CO})_6]^{2\ominus}$ (C) $[\text{Mn}(\text{CO})_6]^\oplus$ (D) CO
5. Identify the correct option :
- (A) $\text{Na}[\text{PtBrCl}(\text{NH}_3)(\text{NO}_2)]$
IUPAC-Sodium amminebromidochloridonitrito-N-platinate(I)
- (B) $[\text{Co}(\text{en})_3]\text{Cl}_3$
IUPAC - tris(ethane-1,2-diamine)cobalt(III) chloride.
- (C) $[\text{PtCl}(\text{NH}_2\text{Me})(\text{NH}_3)_2]\text{Cl}$
IUPAC -diamminechlorido(methylamine)platinum(II) chloride
- (D) $\text{Fe}(\pi\text{-C}_5\text{H}_5)_2$
IUPAC-Bis(η^5 -cyclopentadienyl)iron(II)

Space for Rough Work

6. Choose correct statement/s.

- (A) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ both are ionization isomers of each other and in both complex all S-O bond length are identical
- (B) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ are hydrate isomers of each other and both isomers can be distinguished by conductivity measurements
- (C) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ can show ionization isomerism as well as linkage isomerism
- (D) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ and $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ are polymerization isomer of each other and both show electrical conductance in aqueous solution.
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

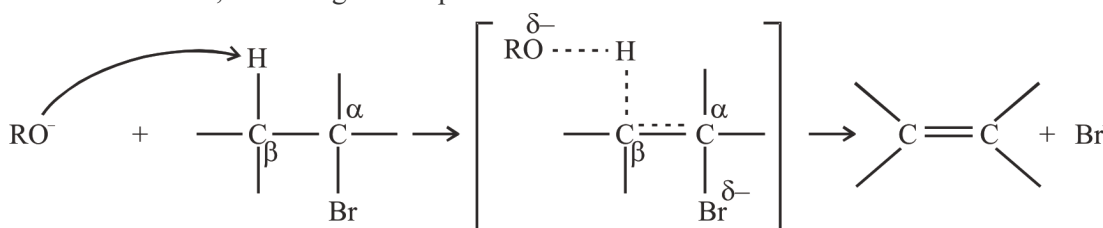
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

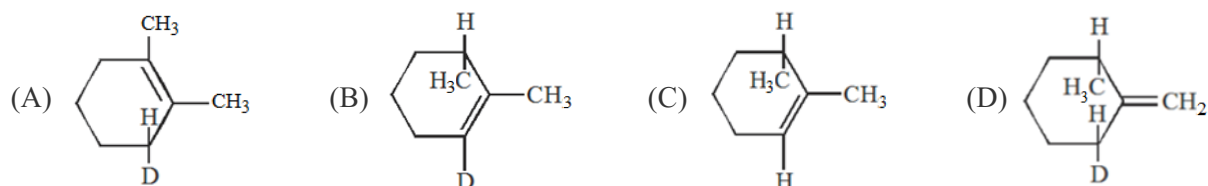
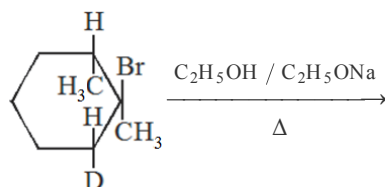
In an E2 reaction, following one step mechanism is involved.



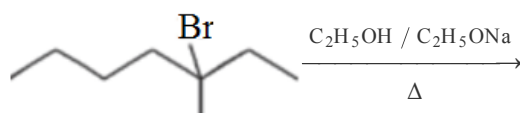
Transition state

As shown in the above mechanism, a β -proton anti to leaving group is abstracted by the alkoxide base. If a less bulky base is used, β -proton is eliminated giving most substituted, most stable product as the major product. On the otherhand if a bulky base is used, β -proton is abstracted from β -carbon where there is least steric hindrance and this gives least substituted, although least stable, alkene as the major product.

7. What is the major product in the following elimination reaction ?



8. In the following reaction, how many elimination products would be formed in principle by E2 mechanism ?



- (A) 2 (B) 3 (C) 4 (D) 5

Space for Rough Work

Paragraph for Questions 9 and 10

Two liquids A and B mix to form an ideal binary liquid solution. On adding solid non-volatile solute C to the solution, A starts polymerising into a hexamer soluble solid 'A₆' (non-electrolyte) following

zero-order kinetics with a rate constant of reaction = ($\frac{10^{-1}}{6}$ moles/min.) If initially 10 moles of A and 20 moles of B were taken to form liquid solution and 5 moles of C were added, then answer the following questions based on above information and data given below :

[Data : vapour pressure of pure A = 100 torr, vapour pressure of pure B = 90 torr].

9. The vapour pressure of the solution 1 hour after solid C is added will be given by :
- (A) $\frac{220}{3}$ torr (B) 98 torr (C) $\frac{2220}{29}$ torr (D) $\frac{2220}{24}$ torr
10. Which of the following statement is incorrect regarding the above process ?
- (A) The vapour pressure will keep on decreasing initially on adding solute C.
- (B) The vapour pressure will become constant 100 min after addition of C.
- (C) Two hours after addition of C, the vapour pressure will become 67.5 torr.
- (D) Four hours after addition of C, the vapour pressure will become 73.75 torr.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

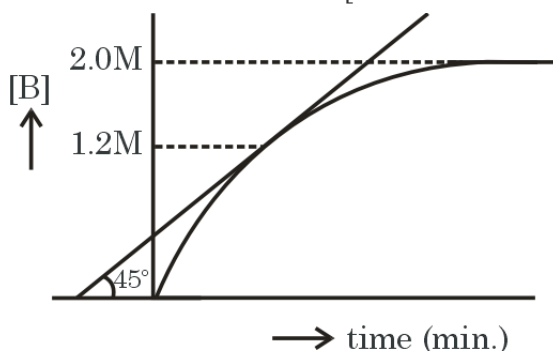
Paragraph for Questions 1 and 2

Heating of the compounds generally loose CO_2 and H_2O . The process is called decarboxylation and dehydration.

1. Which of the following di-carboxylic acid on heating gives cyclic anhydride?
(i) Oxalic acid (ii) Succinic acid (iii) Glutaric acid (iv) Maleic acid
(v) Fumaric acid
2. Which of the following compounds on heating gives cyclic ester?
(i) γ -hydroxy acid (ii) Pimelic acid (iii) Urea (iv) Benzoic acid (v) δ -hydroxy acid

Paragraph for Questions 3 and 4

Following plot is drawn for a first order reaction $\text{A} \rightarrow 2\text{B}$ [Given $\ln 2 = 0.7$]



3. What is the half life of reaction in minutes ?
4. What is the initial rate of reaction (in $\text{M}/\text{min.}$) ?

Space for Rough Work

Paragraph for Questions 5 and 6

According to Werner's coordination theory every central metal atom show two type of valencies in the complexes and try to satisfy both of it's valencies.

5. For complex $\text{CoCl}_3 \cdot 5\text{NH}_3$

If P.V. = x

S.V. = y

and total number of ions in aqueous solution per molecule of complex is = z

then calculate $\frac{z}{x+y} = ?$

6. How many moles of $\text{AgCl}(\text{ppt.})$ is obtained when one mol aqueous solution of $\text{PtCl}_4 \cdot 4\text{NH}_3$ is treated with excess of AgNO_3 solution.

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. The rate constant for decomposition of $\text{COCl}_2(\text{g})$ according to reaction :
 $\text{COCl}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{Cl}_2(\text{g}) ; \Delta H = 19 \text{ kcal mole}^{-1}$
 is expressed as : $\ln k = 15 - \frac{11000}{T(\text{K})}$
 Calculate activation energy for formation of COCl_2 from $\text{CO}(\text{g})$ & $\text{Cl}_2(\text{g})$ in kcal mole^{-1} ?
8. How many given ligands can act as π -electron donor and π^* -acceptor in complexes
 $\text{CH}_2 = \text{CH}_2$, NH_3 , PF_3 , C_6H_6 , C_5H_5^- , H_2O
9. What is the difference between the EAN value of $[\text{Fe}(\text{CO})_5]$ & $[\text{Fe}(\text{CO})_2(\text{NO})_2]$

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

- Tangent from $\left(0, -\frac{1}{2}\right)$ to $y^2 = x^3$ passes through -
(A) (4, 8) (B) (1, -1) (C) (1, 1) (D) (3, 4)
- Consider the function $f(x) = \cos^{-1}([e^x] - 1) + \sin^{-1}([e^x])$, then
[Note : $[.]$ denotes greatest integer function]
(A) the domain of $f(x)$ is $(-\infty, \ln 2]$ (B) the domain of $f(x)$ is $(-\infty, \ln 2)$
(C) the range of $f(x)$ is a singleton set (D) it is many-one function
- If $\lim_{x \rightarrow 0} \frac{x^n - (\sin x)^n}{x^m} = L$ (finite), where $m, n \in \mathbb{N}$, then
(A) If $m - n = 2$, then $L = \frac{n}{6}$ (B) If $m + n = 2$, then $L = \frac{n}{6}$
(C) If $m - n > 2$, then L does not exist (D) If $m - n < 2$, then L does not exist

Space for Rough Work

4. Let $f(x)$ be an even periodic function with period 4 defined partially as $f(x) = \begin{cases} e^x & x \in [0, 1) \\ e(2-x) & x \in [1, 2] \end{cases}$.

In the interval $[-10, 10]$, identify the correct option(s) -

- (A) $f(x)$ is non-derivable at exactly 14 points (B) $f(x)$ is non-derivable at exactly 19 points
(C) Range of $f(x)$ is $[0, e]$ (D) $f(x)$ is discontinuous at exactly 8 points

5. If Lagrange's mean value theorem is applicable to the function $f(x) = \begin{cases} 3, & x = 0 \\ -x^2 + a, & 0 < x < 1 \\ mx + b, & 1 \leq x \leq 2 \end{cases}$ for $x \in [0, 2]$,

then check the correct option-

- (A) $a = 3$ (B) $b = 0$ (C) $m = -2$ (D) $a + b + m = 5$

6. Let $f(x) = \begin{cases} (x+1)e^x & x \leq 0 \\ \frac{x+2\alpha}{2} & 0 < x < 2 \\ x^2 - 2 & x \geq 2 \end{cases}$, then identify the correct statement(s)-

- (A) $x = 0$ is point of local maxima of $f(x)$ for $\alpha < 1$
(B) $x = 2$ is point of local minima of $f(x)$ for $\alpha > 1$
(C) $f(x)$ is increasing at $x = 0$ for $\alpha \geq 1$
(D) $f(x)$ is increasing at $x = 2$ for $\alpha \leq 1$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Given a function f differentiable at $x = 1$ such that $f'(1) = 3$ and satisfying $f(xy) = f(x) + f(y) \forall x, y > 0$.

7. Consider the following statements :

- (I) $f(x)$ is discontinuous at exactly one point in $\mathbb{R}^+$
- (II) $f(x)$ is not differentiable at exactly one point in $\mathbb{R}^+$
- (III) $f(x)$ is differentiable $\forall x \in \mathbb{R}^+$

Identify the correct option(s) -

- (A) Only statement-I is true
- (B) Only statement-II is true
- (C) Only statement-III is true
- (D) Both statement-I & II are true

8. Which of the following is true for $y = |f(|x|)|$ -

- (A) non derivable at exactly 3 points in its domain
- (B) non derivable at exactly 2 points in its domain
- (C) discontinuous at exactly 2 points in its domain
- (D) continuous and derivable throughout its domain

Space for Rough Work

Paragraph for Questions 9 and 10

Consider $f : [0, \infty) \rightarrow [0, \infty)$; such that $f(x)$ is bijective and satisfies $f(x)e^{f(x)} = x$.

9. The greatest integer less than or equal to the value of $f(f(e^{e+1}))$, is -

(A) 0 (B) 1 (C) 2 (D) 3

10. The value of $\lim_{x \rightarrow \infty} \frac{f(x)}{\ln x}$, is -

(A) 0 (B) 1 (C) e (D) non-existent

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

$$\text{Let } f(x) = \begin{cases} (2a-1) \sin^{-1} \left(\frac{2x+3}{4} \right) & , \quad x \in \left[-\frac{7}{2}, 0 \right) \\ b \cos^{-1} \left(\frac{\sqrt{7}+x}{4} \right) & , \quad x \in \left[0, 4-\sqrt{7} \right], \\ b \tan^{-1} \left(\sqrt{7}+x \right) - 2 \sin^{-1} \frac{4}{\sqrt{17}} & , \quad x > 4-\sqrt{7} \end{cases}$$

is a continuous function in $\left[-\frac{7}{2}, \infty \right)$. Then answer the following question

1. $a + b$ is equal to
2. Number of real solutions of $f(x) = -\frac{4\pi}{3}$ is

Paragraph for Questions 3 and 4

Consider $f(x) = x^3 - 6x^2 + 9x + 1$.

3. If Rolle's theorem holds in $[0, a]$, then positive integral value of a is
4. Point of inflection of $f(x)$ is

Space for Rough Work

Paragraph for Questions 5 and 6

If $f(x) = \cos^5 x$ and $g(x) = \sin x$.

5. If $\int \frac{f(x)}{g(x)} dx = \ell \log |\sin x| + m \sin^4 x + n \sin^2 x + C$, (where C is constant of integration), then $\frac{\ell - n}{m}$ is
6. If period of $f(x) - g(x)$ is $\alpha\pi$, then α is

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Consider the equations

$$x - y + z = 1$$

$$\lambda x + 3y + 2z = 3$$

$$3x + \lambda y + z = 2$$

If equations has unique solutions such that $\lambda \neq k_1$ and $\lambda \neq k_2$, where $k_1, k_2 \in \mathbb{I}$ and $|k_1| > |k_2|$ then $4 \left| \frac{k_2}{k_1} \right|$ is equal to

8. Water is poured into an inverted conical vessel of radius 1 m & height 2 m, at the rate of 23.1 litre/min. The rate at which the water level is rising at the instant when the depth is 70cm is equal to (in cm/min.)

(Use $\pi = \frac{22}{7}$)

9. Number of critical points of $f(x) = \frac{\ln x}{1+x} - x, x \geq 1$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work